STRUCTURAL ANALYSIS AND STEEL DESIGN OF A G+9 RC RESIDENTIAL BUILDING

Software: ETABS v20.0.0
Design Standard: ACI 318-05 / BNBC 2020

PROJECT PORTFOLIO REPORT

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Hypothetical Academic Portfolio Project

Purpose: Structural engineering learning and portfolio demonstration

1. Introduction

This report presents the structural modeling, analysis and design review of a hypothetical G+9 reinforced concrete (RC) residential building using ETABS. The project has been developed for educational purposes and to demonstrate structural engineering skills in a professional portfolio.

The study considers the behavior of the building under gravity and lateral loading. The structural model includes reinforced concrete beams, columns, floor systems and other relevant structural components. Static and dynamic analysis procedures are used to investigate the structural response, including displacement, storey drift, modal behavior and seismic response. The project also includes a review of selected concrete frame design results and structural response parameters. The findings are intended to demonstrate the application of structural analysis software and engineering judgment. As a hypothetical academic project, the model and its results should not be interpreted as approval for actual construction.

2. Objectives

The main objective of this project is to develop and evaluate a hypothetical G+9 reinforced concrete residential building using ETABS. The project aims to demonstrate the practical application of structural modeling, analysis and design-review procedures.
The specific objectives are:

- To develop a three-dimensional structural model of the building using ETABS.
- To define the material properties, structural sections, supports and floor systems used in the model.
- To assign gravity and lateral loads and establish the relevant load combinations.
- To perform structural analysis and examine the resulting structural response.
- To review storey displacements, storey drifts, modal properties and response-spectrum results.
- To review selected reinforced concrete beam and column design results.
- To organize the modeling, analysis and design-review outputs into a technical report suitable for an engineering portfolio.

The project also aims to strengthen the understanding of structural behavior and the importance of verifying software inputs and outputs through engineering judgment.

3. Building Description

The project considers a hypothetical G+9 reinforced concrete residential building modeled in ETABS. The building consists of a ground floor and nine upper storeys with the structural system represented by reinforced concrete framing elements and floor systems.

A three-dimensional analytical model was developed to represent the building's structural configuration and evaluate its response to the applied loading conditions. The model includes structural members, material definitions, support conditions, load patterns and load combinations.

The structural response was investigated under gravity and lateral loading conditions. Dynamic characteristics were also evaluated through modal analysis while response-spectrum cases were included to study the building's seismic response.

The building configuration and modeling assumptions are intended for educational and portfolio purposes. The project does not represent a verified real-world building and its suitability for actual construction has not been established.

4. Software Used

Software

ETABS was used to develop the three-dimensional structural model, define loading conditions, perform structural analysis and review selected reinforced concrete frame design results. The software provides analytical results for structural displacements, storey drifts, modal properties, base reactions and member design checks based on the model inputs and selected analysis and design settings.

Analysis and Design Basis

The project includes gravity-load analysis, lateral-load analysis, modal analysis using the Eigen vector method and response-spectrum analysis. The selected load cases and combinations were defined in ETABS to investigate the structural response under different loading conditions.

The concrete frame design settings and applicable design provisions must be distinguished from the requirements of the Bangladesh National Building Code (BNBC) 2020. Although BNBC 2020 is relevant to the intended context of this project, full compliance has not been established merely by completing analysis and design checks in ETABS. The selected code settings, seismic parameters, load combinations and applicable acceptance criteria require verification before any compliance claim can be made.

This project is hypothetical and intended for educational and portfolio demonstration. The analysis results depend on the assumptions, input data and software settings adopted in the model.

5. Structural Modeling

5.1 Model Development

A three-dimensional analytical model of the hypothetical G+9 reinforced concrete residential building was developed using ETABS. The model represents the building's structural configuration through analytical elements, material properties, section properties, support conditions and floor systems.

The structural model was prepared to investigate the behavior of the building under gravity and lateral loading. Appropriate load patterns, analysis cases and load combinations were defined for the intended structural evaluation.

5.2 Structural Elements and Boundary Conditions

Reinforced concrete beams and columns were represented using frame elements. The floor system and other relevant structural components were modeled according to the adopted modeling assumptions. Support conditions and structural connectivity were defined to represent the intended behavior of the analytical model.

The accuracy of the analytical results depends on the adequacy of the element properties, connectivity, boundary conditions, stiffness assumptions and load assignments.

5.3 Analysis Model Verification

After completing the model, structural analysis was performed in ETABS. The analysis completed without software warnings. However, the absence of warnings alone does not establish that the model is free from errors. Engineering verification of the geometry, assignments, loading, boundary conditions and analytical results remains necessary.

6. Material Properties and Section Definitions

6.1 Material Properties

Material properties were defined in ETABS for the reinforced concrete structural system. These properties influence the stiffness, strength, deformation and analytical response of the building. The material grades and associated parameters should be reported according to the values actually assigned in the model. These values should not be assumed to represent verified construction materials because the project is hypothetical.

6.2 Section Properties

Section properties were assigned to the structural members to represent their geometric characteristics and analytical stiffness. Beam and column sections form the primary components of the modeled reinforced concrete frame.

The assigned dimensions and reinforcement-design settings influence the analysis results and member design checks. The final report should document the actual section dimensions and material properties used in ETABS rather than introducing unverified values.

6.3 Design Implications

The suitability of the selected sections depends on strength, stiffness, stability, serviceability and applicable design requirements. A successful software design run should therefore be treated as one part of the structural evaluation rather than conclusive evidence of overall code compliance.

7. Loading and Load Combinations

7.1 Load Patterns

Several load patterns were defined in ETABS to represent the anticipated actions on the hypothetical building. These included gravity loads, lateral seismic loads and wind loads.

The model included the following load patterns:

- Dead load (Dead)
- Floor live load (LLf)
- Roof live load (LLr)
- Floor finish load
- Partition wall load
- Lift load
- Seismic loads in the global X and Y directions (Ex and Ey)
- Wind loads in the global X and Y directions (Wx and Wy)

The load patterns were used to represent the different actions considered in the structural analysis. Their magnitudes, directions, and self-weight settings should be consistent with the assumptions and requirements adopted for the project.

7.2 Load Cases

Static load cases were used to investigate the response under the assigned static loading conditions. Modal analysis was performed using the Eigen vector method to evaluate the building's natural vibration characteristics.

Response-spectrum load cases were also defined in the global X and Y directions. These cases used the modal analysis results to evaluate the building's dynamic response to the specified response spectrum.

7.3 Load Combinations

Load combinations were defined to evaluate structural response under different combinations of applied actions. The combinations selected for design and analysis should reflect the intended limit states and applicable design provisions.

The load factors and combination rules require verification against the governing design basis. The existence of load combinations in ETABS does not, by itself, establish compliance with BNBC 2020 or any other design standard.

8. Structural Analysis

8.1 Linear Static Analysis

Linear static analysis was used to evaluate the building's response under selected static load cases. The resulting forces, displacements, reactions and storey drifts provide information about the modeled structure's response to the assigned loading.

8.2 Modal Analysis

Modal analysis was conducted using the Eigen vector method with 16 modes. The analysis provided the natural periods, mode shapes and modal participating mass ratios used to understand the building's dynamic characteristics.

The modal results were also used in the response-spectrum analysis cases. The adequacy of the selected number of modes was assessed in relation to the cumulative participating mass in the principal horizontal directions.

8.3 Response-Spectrum Analysis

Response-spectrum analysis was performed in the global X and Y directions using the defined response-spectrum function and modal results. The analysis produced estimates of the building's seismic response, including horizontal base reactions and deformation-related results.

The response-spectrum results must be reviewed alongside the adopted seismic parameters, spectrum definition, modal combination method and applicable base-shear scaling requirements.

8.4 Review of Analytical Results

The analytical results were reviewed through selected output tables and graphical displays in ETABS. These included deformed shapes, modal behavior, base reactions, storey drifts, modal participating mass ratios and reinforced concrete frame design results.

The review focused on understanding structural behavior and identifying results requiring further verification. Numerical results were not treated as proof of code compliance without checking the relevant acceptance criteria.

9. Analysis Results and Review

9.1 Overview of Analysis Results

The structural analysis results were reviewed to understand the response of the hypothetical G+9 reinforced concrete residential building under the modeled loading conditions. The review included modal properties, storey drift, response-spectrum base reactions and selected reinforced concrete frame design results.

The results represent the behavior of the analytical model under the defined assumptions and input parameters. Their engineering significance depends on the accuracy of the model and verification against applicable design criteria.

9.2 Modal Analysis Results

Modal analysis was performed using the Eigen vector method with 16 modes. The first three modes exhibited predominantly torsional, Y-direction translational and X-direction translational behavior respectively.

The cumulative participating mass ratios reached 95.66% in the global X direction and 96.35% in the global Y direction by Mode 16. These results indicate that the selected modes captured substantial portions of the modeled building's translational mass participation in both principal horizontal directions.

The modal periods and mode shapes provide insight into the dynamic characteristics of the building. Further evaluation of torsional behavior and the applicable modal-participation requirements is necessary before finalizing the seismic assessment.

9.3 Storey Drift Results

Storey drift results were reviewed for the Ex and Ey static seismic load cases.

The maximum reported storey drift ratios were:

- Ex: 0.001341, at Story 4.
- Ey: 0.001763, at Story 4.

The maximum reported drift ratio for Ey was approximately 31% greater than that for Ex. This indicates that the modeled response was not identical in the two principal horizontal directions.

Storey drift is an important serviceability and seismic-performance parameter. The reported values must be checked against the applicable code limits using the correct drift definition, load-case requirements and any required amplification factors. These results alone do not establish code compliance.

9.4 Response-Spectrum Base Reactions

The latest recorded response-spectrum base-reaction results are summarized below.

Response-spectrum case	Horizontal direction	Base shear
RSx	Global X	483.050 kip
RSy	Global Y	440.239 kip

The RSx case produced a dominant horizontal base shear of 483.050 kip in the global X direction, while RSy produced 440.239 kip in the global Y direction.

These values represent the latest recorded analysis results and should be confirmed against the final saved model and response-spectrum scale factors. The applicable minimum base-shear requirements and scaling procedures must be verified before drawing conclusions about seismic design adequacy.

9.5 General Interpretation

The analysis outputs provide information about the building's dynamic behavior and response to the assigned loading conditions. The modal results indicate substantial cumulative horizontal

mass participation, while the storey drift results show a greater maximum drift ratio in the Y direction than in the X direction.

The response-spectrum base reactions provide a basis for further seismic verification. However, a complete structural assessment also requires verification of the model assumptions, load definitions, seismic parameters, member design results and applicable acceptance criteria.

10. Structural Design Checks

10.1 Reinforced Concrete Beam Design

Concrete frame design checks were performed in ETABS and beam design summary results were reviewed. The results provide information about the design demands and calculated reinforcement requirements for the modeled beams.

The beam design results should be assessed for flexural strength, shear strength, reinforcement detailing and applicable minimum and maximum reinforcement requirements. The final acceptance of each member depends on the governing design provisions and the correctness of the model inputs.

10.2 Reinforced Concrete Column Design

Concrete column design results were reviewed using the ETABS design output tables. Column adequacy depends on the combined effects of axial force and bending moments as well as shear, slenderness, reinforcement detailing and other applicable design requirements.

The reported design outputs require careful interpretation. A graphical display or an incomplete summary table should not be treated as conclusive evidence that every column satisfies all design requirements.

10.3 Shear and Flexural Results

Beam and column shear-envelope and beam flexural-envelope results were exported for review. These results support the investigation of internal-force demands under the selected load combinations.

The governing demands should be compared with the corresponding design capacities and detailing requirements. The exported tables form part of the supporting project documentation but do not independently establish full structural adequacy.

10.4 Design Review Summary

ETABS completed the selected concrete frame design procedures and relevant design output tables were collected for documentation. The results provide a basis for reviewing member behavior and identifying areas requiring further verification.

The project report does not claim full compliance with BNBC 2020 because the governing design settings and all applicable acceptance criteria have not been independently verified.

11. Limitations of the Project

This project is a hypothetical academic portfolio exercise and does not represent a commissioned building design. Its purpose is to demonstrate structural modeling, analysis and design-review skills using ETABS.

The principal limitations are as follows:

- The building configuration, material properties, section properties and loading assumptions are those adopted for the analytical model and have not been independently validated for a real site.
- The analysis results depend on the accuracy of the model geometry, member assignments, support conditions, stiffness assumptions and load definitions.
- The seismic parameters, response-spectrum definition, load combinations and base-shear scaling procedures require verification against the applicable design requirements.
- Completion of software analysis and design procedures does not, by itself, establish compliance with all relevant provisions of BNBC 2020.
- A complete real-world design would require additional verification of structural detailing, foundations, geotechnical conditions, construction requirements and other applicable engineering considerations.

Accordingly, the results should be interpreted as an educational demonstration rather than a construction-ready design or professional certification of structural safety.

12. Conclusion

A hypothetical G+9 reinforced concrete residential building was modeled and analyzed using ETABS as part of a structural engineering portfolio project. The work included the definition of structural elements and material properties, assignment of gravity and lateral loading, establishment of load cases and combinations and execution of structural analysis and selected concrete frame design procedures.

The modal analysis provided insight into the building's dynamic characteristics with cumulative participating mass ratios of 95.66% in the global X direction and 96.35% in the global Y direction by Mode 16. The reported maximum static seismic storey drift ratios were 0.001341 for Ex and 0.001763 for Ey. The latest recorded response-spectrum base shears were 483.050 kip in the X direction and 440.239 kip in the Y direction.

The project demonstrates the use of ETABS for structural modeling, analysis, interpretation of analytical results and documentation of selected design outputs. It also emphasizes the importance of verifying model assumptions, code settings, seismic parameters and member acceptance criteria before making conclusions about structural adequacy.

Because this is a hypothetical portfolio project, the results do not constitute approval for construction or confirmation of full BNBC 2020 compliance. Further engineering verification would be required before applying the model or its results to an actual building.